Topic 1.5 – Atomic structure and the periodic table	
The role of instrumentation in analytical chemistry is illustrated by mass spectrometry. *The study of atomic structure gives some insight into the types of evidence which scientists use to study electrons in atoms. This leads to an appreciation of one of the central features of chemistry which is the explanation of the properties of elements and of the patterns in the periodic table in terms of atomic structure.*	
1.5b	Students could consider the use of the mass spectrometer by the pharmaceutical industry to produce data concerning molecular mass. **HSW 3**
1.5e	Looking at electronic structure can provide an opportunity to discuss the historical development of a model and the way that evidence is used to generate a chemical model that is then modified as more data becomes available. **HSW 1, HSW 7**
1.5f	Students should consider why a more sophisticated model is needed to explain the properties of electrons in atoms and the way atoms join together. **HSW 1, HSW 7**
1.5g	Students could discuss the idea that trying to remember the electronic structure of every element is difficult so chemists devised a way to rationalise the information about the electronic structure of each atom into a pattern that could help to predict the electronic structure of other atoms. Students need to realise that the idea helps chemists decide on the minimum energy arrangement of a particular number of electrons around a nucleus and that this has limitations. **HSW 1**